

- **Collaboration tools:** Generative AI can create intelligent collaboration tools that facilitate communication and knowledge sharing among internal users, fostering a more connected and productive work environment.

## Developers

Developers play a crucial role in creating, maintaining, and enhancing a company's software applications. Generative AI can support developers by automating code generation, optimising software design, and enhancing testing processes. Ethical solutions have been a top priority for organisation leaders. Bias mitigation is a very important part of the everyday AI development journey.

- **Code generation:** Generative AI can automate the creation of code snippets or entire applications based on developers' specifications, accelerating development cycles and improving productivity.

- **Software design optimisation:** By analysing existing codebases and design patterns, generative AI can provide developers with recommendations for optimising software architecture and improving overall performance.
- **Enhanced testing:** Generative AI can create intelligent testing tools that automatically generate test cases, identify bugs, and ensure the quality of software applications.

## Technology operations

Technology operations teams are responsible for managing and maintaining a company's IT infrastructure. Generative AI can assist these teams by automating infrastructure management, predicting system failures, and optimising resource allocation.

- **Infrastructure management automation:** Generative AI can automate routine infrastructure management tasks, such as

server provisioning and network configuration, enabling technology operations teams to focus on more strategic initiatives.

- **Predictive maintenance:** By analysing system logs and performance metrics, generative AI can predict potential system failures and alert technology operations teams to proactively address them.
- **Resource allocation optimisation:** Generative AI can analyse patterns in resource usage and provide recommendations for optimising resource allocation, ensuring maximum efficiency and cost savings.

## Innovation in open source leading to GenAI adoption

In the realm of generative AI programming, open source tools have made it possible for end users, internal users, developers, and technology operations teams to access cutting-edge technologies and innovate faster than

| Persona       | Use case                             | Adoption needs                                                    | Open source tools         |
|---------------|--------------------------------------|-------------------------------------------------------------------|---------------------------|
| End user      | Personalised content generation      | Integration with existing systems, user data analysis             | TextGenRNN, GPT-2         |
| End user      | Predictive analytics                 | Data mining, pattern recognition, predictive modelling            | Prophet                   |
| End user      | Enhanced customer support            | Chatbot development, natural language processing                  | Rasa, Botpress            |
| Internal user | Task automation                      | Workflow analysis, automation tools integration                   | Automatica, TagUI         |
| Internal user | Data-driven insights                 | Data analytics, visualisation tools, business intelligence        | Pandas, NumPy, Matplotlib |
| Internal user | Collaboration tools                  | Integration with communication platforms, AI-driven features      | Mattermost, Rocket.Chat   |
| Developer     | Code generation                      | AI-powered code generation tools, integration with IDEs           | Codota, Kite              |
| Developer     | Software design optimisation         | Code analysis, design pattern recognition, optimisation tools     | SonarQube, CodeClimate    |
| Developer     | Enhanced testing                     | AI-driven testing tools, test case generation, bug detection      | Diffblue, EvoSuite        |
| Operations    | Infrastructure management automation | Automation tools, integration with IT infrastructure              | Ansible, Terraform        |
| Operations    | Predictive maintenance               | System log analysis, performance monitoring, predictive modelling | Graylog, Prometheus       |
| Operations    | Resource allocation optimisation     | Resource usage analysis, optimisation algorithms                  | Kubernetes, Apache Mesos  |

Table 1: Mapping adoption needs of use cases to open source tools